System Limitations, Threat Boundaries & R&D;

Honest Engineering Boundary Analyses, Coercion Threat Models, Platform Sandbox Constraints & Active Research Register • Final MVP

1. Physical Coercion & Adversary Threat Boundaries

Standard mobile security threat models assume a remote adversary attempting to breach a device across a network. In Gender-Based Violence (GBV) and intimate partner violence contexts, the threat reality is fundamentally inverted: the adversary has **physical possession, continuous proximity, and the power of coercion** over the survivor. Honest engineering requires recognizing the absolute boundary of what client software can and cannot defend against:

PHYSICAL THREAT BOUNDARY REALITY

Engineering Boundary Principle: No mobile application can completely prevent detection if an abusive partner installs commercial stalkerware/keyloggers at the OS kernel level, or uses physical violence to force a survivor to unlock their device. Umkhuseli designs around this boundary through **plausible deniability, multi-platform zero-trace PWAs, and authentic decoy utilities**.

Threat Vector & Boundary	Inherent Technical Limitation	Umkhuseli Architectural Countermeasure
Intense Device Inspection	App stores (Google Play, Apple App Store) maintain unalterable download histories and purchase records.	**Zero-Install Vercel Web PWA:** Survivors facing high scrutiny access the portal in incognito mode. Bypasses app stores entirely, leaving 0 install records and 0 battery footprint.
Forced / Duress Unlocking	Under physical duress, survivors cannot refuse to unlock their smartphone without risking violence.	**Duress PIN 9999:** Entering 9999 generates a realistic municipal voucher error receipt ('STS Error: Communication Fault') while silently dispatching the background SOS beacon.
Surprise Over-the-Shoulder Discovery	Adversary enters the room while survivor has the evidence vault or TCC navigator open.	**0-ms Camouflage Scrub:** Back button or panic tap instantly wipes decrypted RAM memory in <10ms and renders the innocent PowerCalc SA municipal calculator.

2. OS Sandboxing & Hardware Constraints

Mobile operating systems impose stringent sandboxing and battery constraints that directly impact emergency reliability:

- **Android 14+ Doze Mode & Aggressive OEM Battery Killers:** Samsung OneUI, Xiaomi HyperOS, and Huawei aggressively throttle background GPS and network threads after 15 minutes of screen-off time. **Mitigation:** Umkhuseli configures an Android 14 foregroundServiceType='location' with an innocuous sticky notification ('PowerCalc SA Tariff Monitor'), requests battery optimization exemptions, and falls back to coarse cell tower network triangulation if GPS hardware sleeps.
- **Budget Hardware Cryptography Degradation:** Many vulnerable users in low-income South African communities rely on entry-level Android devices (MediaTek Helio A22, Unisoc SC9863A) lacking discrete Secure Elements. Enabling setIsStrongBoxBacked(true) causes StrongBoxUnavailableException. **Mitigation:** Wrapped in a robust try-catch handler that gracefully degrades to the device's hardware-backed Trusted Execution Environment (TEE) AES-256-GCM.

- **Samsung USSD '#' Stripping Bug:** Samsung Android 14 dialers treat the trailing '#' in *120*7867# as a URI fragment and strip it. **Mitigation:** Explicitly wrapped in Uri.encode(ussdString) to ensure reliable zero-airtime USSD callback execution.

3. South African Infrastructure Realities & Mitigations

- **Stage 6 Loadshedding Base Station Blackouts:** During 4-6 hour rotational power cuts, cellular base station backup batteries become exhausted, causing complete LTE data drops. **Mitigation:** Local-first Room SQLite and Dexter IndexedDB architecture allows complete offline logging. Beacons queue in an offline outbox and switch to SMS/GPS binary packets.
- **Data Poverty & Airtime Depletion:** Prepaid airtime balances are frequently R0.00. **Mitigation:** Direct integration with South Africa's statutory zero-rated infrastructure: SAPS 10111, GBV Command Centre 0800 428 428, and USSD *120*7867# callback pings.

4. Legal Admissibility Realism & Ethical Guardrails

Umkhuseli maintains rigorous intellectual honesty regarding legal outcomes:

- **ECTA Section 15 Alignment vs 'Guaranteed Admissibility':** No cryptographic system can guarantee court admissibility in South Africa. Magistrates and High Court judges evaluate evidence based on relevance and probative value. Umkhuseli creates **tamper-evident contemporaneous records** with SHA-256 hash chains that satisfy Section 15 of ECTA (Act 25 of 2002) to corroborate sworn affidavits and J88 clinical notes.
- **Ethical AI Guardrails:** The Form 2 drafting assistant does NOT generate binding legal decrees. It functions as a structured intake assistant aiding court clerks and social workers, with mandatory disclaimers advising review by Legal Aid SA.

5. Active Research & Development (R&D;) Register

R&D; Project Track	Research Objective & Technology	Target Horizon
Curve25519 OPRF PSI	Oblivious Pseudorandom Function Private Set Intersection for contact matching immune to South African MSISDN rainbow tables.	Phase 3 (Horizon 2)
TinyML Acoustic Classifier	8-bit quantized on-device audio model recognizing acoustic distress and glass breaks within 15-second RAM buffers.	Phase 2 (Horizon 1)
PostGIS Ward Safety Mesh	Zero-knowledge municipal ward polygon geofencing for pseudonymous peer safety alerts.	Phase 3 (Horizon 2)